

CSE2218: DSA2 Lab
Practice Question

- Write a **C++ program** to define a class **“Product”** with the following attributes and functions:
Attributes:
 - name (string : private)
 - category (string : private)
 - price (double : public)
 - brand (string : private)
- **Define necessary constructor and getter/setter methods.**
- Define a function **“void ComputeFinalPrice(double percentageDiscount)”** inside the class that calculates the **final price after applying discount** using the following rule:
$$\text{price} = \text{price} - \text{price} * \text{percentageDiscount};$$
- Define a function **“void display()”** inside the class that displays the **name, category, brand, and final price** of the product.
- **In the main() function**
 - Create a **vector of Product objects** and insert the following products:
 - "Laptop", "Electronics", 85000.00, "Dell"
 - "Mobile", "Electronics", 45000.00, "Samsung"
 - "Chair", "Furniture", 5000.00, "Otobi"
 - "Table", "Furniture", 8000.00, "Navana"
 - "Headphone", "Electronics", 3000.00, "Sony"
- The shop is giving **15% discount on Electronics** product and **10% on all other** products. Now display the products after applying discount.
- **Sort the products based on price in ascending order.**
- **Sort the products based on category in descending order** and display all products.